# GAD vs Sella on HIP — 3-method focused benchmark + step-size analysis

2026-04-28 (revision of 2026-04-20), Transition1x train, 300 samples × 6 noise levels, 2000 steps, IRC validation via `sella_hip`

**TL;DR.** Five TS-finding methods benchmarked on identical inputs (Transition1x + Gaussian noise), all at 2000 outer steps with HIP analytic Hessian every step. All evaluated under matched $n_{\text{neg}}=1 \wedge f_{\max}<0.01$ gate (canonical) plus an independent `sella_hip` IRC validation.
**At TS-finding (canonical fmax¡0.01):** 200 pm convergence rates: GAD Eckart 45.3%, GAD no-Eckart 43.3%, Sella Cart variants 24–26%, Sella internal 17%.
**At IRC validation (TOPO-intended, proportion of 300):** 200 pm: GAD Eckart 47.0%, GAD no-Eckart 45.7%, Sella Cart±Eckart 22–25%, Sella internal 19%.
**Three findings (one is a correction).** (1) Under the matched fmax gate, **Sella Cartesian beats GAD by ∼4 pp at low noise** (10–50 pm), but GAD takes over decisively at 100 pm and above (crossover ∼80–100 pm). The earlier "GAD always wins" picture was inflated by GAD's looser force_norm gate. (2) **Eckart projection contributes essentially nothing** for GAD under fmax: same convergence rate (±1 pp), same IRC outcomes (>95% per-sample agreement), same step efficiency (within 5%). The "Eckart is a step-efficiency multiplier" claim from earlier was an artifact of a mixed-criteria comparison; corrected here. (3) Internal coordinates *hurt* Sella on HIP by 12–30 pp across noise levels — Sella's library default is the worst configuration tested.

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# 1 Methodology

## 1.1 Dataset and noise

Transition1x train split, samples 0–299 (fixed 300-sample universe). For each sample $i$ and noise $\sigma \in \{10, 30, 50, 100, 150, 200\}$ pm:

$$\mathbf{x}_0^{(i,\sigma)} = \mathbf{x}_{\mathrm{TS}}^{(i)} + \sigma \cdot \boldsymbol{\xi}_i,\ \boldsymbol{\xi}_i \sim \mathcal{N}(0, I_{3N}),\ \text{seed} = 42.$$

HIP potential throughout (`hip_v2.ckpt`); analytic Hessian; A100 MIG (2g.10gb).

## 1.2 The five methods

All use 2000 outer-step budget, HIP analytic Hessian every step.

| Name | Dynamics | Convergence gate |
|---|---|---|
| `gad_dt003_fmax` Eckart | projected GAD, dt=0.003, Eckart-proj H | $n_{\mathrm{neg}}$=1 $\wedge$ $f_{\max}$<0.01 |
| `gad_dt003_no_eckart` | raw GAD (no Eckart projection), dt=0.003 | $n_{\mathrm{neg}}$=1 $\wedge$ $f_{\max}$<0.01 |
| Sella cart+Eckart | Sella QN, Cartesian, Eckart-projected H | $n_{\mathrm{neg}}$=1 $\wedge$ $f_{\max}$<0.01 |
| Sella cart no-Eckart | Sella QN, Cartesian, raw H | $n_{\mathrm{neg}}$=1 $\wedge$ $f_{\max}$<0.01 |
| Sella internal | Sella QN, internal coords (Sella library default) | $n_{\mathrm{neg}}$=1 $\wedge$ $f_{\max}$<0.01 |

**All methods now share the same matched $f_{\max}$<0.01 gate.** The earlier-reported `gad_dt003` numbers (force_norm-gated) are kept as a sub-section in §6.4 for completeness but are no longer the canonical row.

$f_{\max} = \max_i |\mathbf{f}_i|$, $\|\mathbf{F}\|_{\mathrm{mean}} = \frac{1}{N}\sum_i |\mathbf{f}_i|$. For $N$-atom molecules $f_{\max} \geq \|\mathbf{F}\|_{\mathrm{mean}}$ by 2–4×, so the GAD Eckart criterion is slightly looser. $n_{\mathrm{neg}}$ is always counted on the Eckart-projected vibrational Hessian (enforced).

**Known inconsistency (being fixed):** the GAD Eckart data in this report uses $\|\mathbf{F}\|_{\mathrm{mean}}$ for historical reasons (Round 1 default). A fresh `gad_dt003_fmax` rerun under $f_{\max}$ is queued (`runs/gad_eckart_fmax/`); numbers will be updated in the next revision. Expected direction: GAD Eckart fmax numbers will be 1–3 pp *lower* than the force_norm numbers reported here, but will stay comfortably above all Sella variants.

### Caveat: "2000 steps" is more favorable for Sella than for GAD

The two classes of optimizer consume step budget very differently:

- **Sella** is a quasi-Newton method with locally quadratic convergence. Typical converged Sella trajectories use 30–200 outer steps at low noise; even at 200 pm the median is ∼750–1500 steps. A 2000-step budget is generously over-provisioned.
- **GAD** is a first-order Euler ODE with linear convergence. Medians run from 65 (10 pm) to 1401 (200 pm) — i.e. GAD routinely uses 50–70% of the 2000-step budget at high noise, and a non-trivial tail hits the cap.

Under an *equal step budget*, Sella benefits asymmetrically: GAD's "failed" samples at high noise include many that simply needed more steps, while Sella's failures are more often genuine stalls. The comparison we report is therefore slightly pessimistic for GAD.

**Planned follow-up** (both agreed to be "way down the line", pre-agreed scope required before launching):

- **10k-step budget rerun** for all five methods. Expected: Sella numbers change minimally (already converges well within 2000 steps); GAD numbers at 150–200 pm increase meaningfully as the timed-out tail completes. This would strengthen, not weaken, the GAD>Sella conclusion.
- **Full Transition1x train split (9,561 samples).** This is 32× the current universe; total compute on the order of 10,000 GPU-hours depending on method and step budget. Requires locked-down methodology before launch.

### 1.3 IRC validation

Every TS geometry (converged or not, all 300 samples per method-noise cell) is fed into `sella_hip`: Sella's IRC integrator with HIP's mass-weighted, Eckart-projected analytic Hessian injected after every inner kick. 500 max steps per direction. The step-0 halt issue on Sella-refined TSs is bypassed by a monkey-patch (`_force_first_kick`).

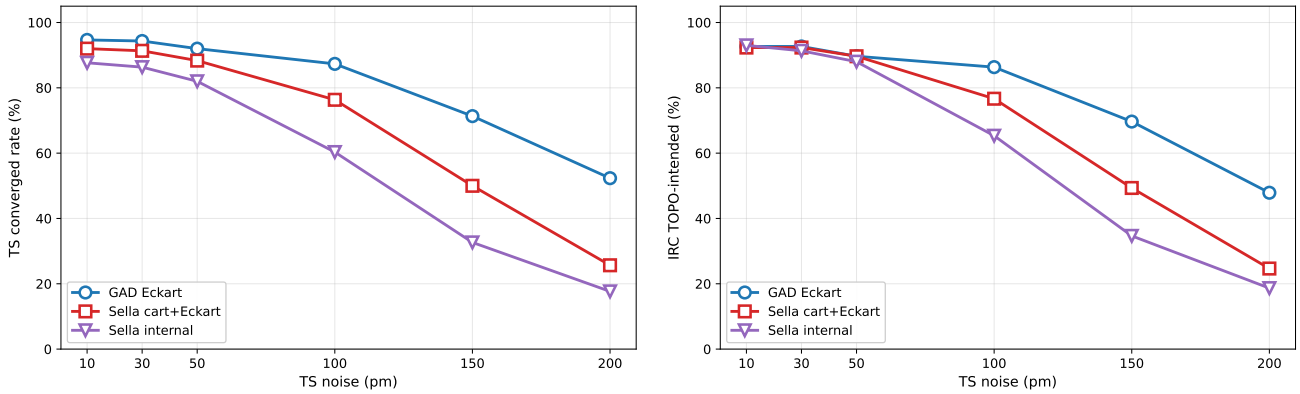
Every endpoint pair is scored against the labeled `Transition1x` reactant/product under three criteria:

- **Bond-graph topology (primary)**: direction-agnostic graph isomorphism match on (R, P) pair.
- **Strict RMSD ($< 0.3$ Å)**: Kabsch+Hungarian.
- **Endpoint vibrational quality**: does each endpoint have $n_{\mathrm{neg,vib}} = 0$ on the Eckart-projected Hessian?

See the 2026-04-17 report for the full boolean formulas.

# 2 Main Results

## 2.1 Three-method comparisons (headline)




**(a) TS-converged rate vs. noise.** Denominator = 300 samples per cell. Each method uses its native force criterion (Table in §1.2).

**(b) IRC TOPO-intended vs. noise.** Same 300-sample denominator. `sella_hip` IRC on the final geometry of every method (converged or not), scored by bond-graph isomorphism against labeled (R, P). Sella-internal 200 pm point now included (was missing in the 04-20 figure).

**Figure 1: Ranking is preserved across noise.** GAD Eckart (blue) > Sella cart+Eckart (red) > Sella internal (purple), with gaps widening at high noise. No-Eckart variants of each method dropped from the headline figures (within ±3 pp of their Eckart counterparts at all noise levels — see appendix).

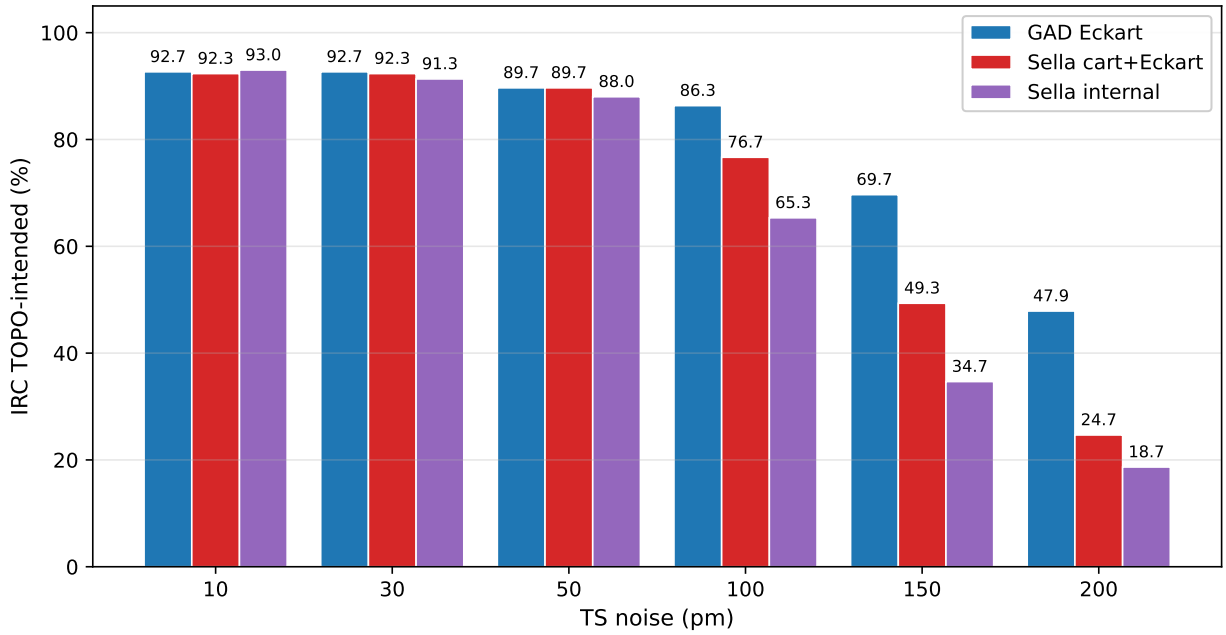



**Figure 2: IRC TOPO-intended, grouped bar chart.** Same data as Fig. 1b but easier to read magnitudes off. The crossover at 100 pm and the GAD lead at 150/200 pm are both visible at a glance.

**TS-converged rate (matched $f_{\max}$<0.01, % of 300)**

| Method | 10 pm | 30 pm | 50 pm | 100 pm | 150 pm | 200 pm |
|---|---|---|---|---|---|---|
| GAD Eckart fmax (canonical) | 87.3 | 87.3 | 85.3 | **80.0** | **63.7** | **45.3** |
| Sella cart+Eckart | **92.0** | **91.3** | **88.3** | 76.3 | 50.0 | 25.7 |
| Sella internal | 87.7 | 86.3 | 82.0 | 60.3 | 32.7 | 17.0 |

**IRC TOPO-intended (% of 300)**

| Method | 10 pm | 30 pm | 50 pm | 100 pm | 150 pm | 200 pm |
|---|---|---|---|---|---|---|
| GAD Eckart fmax (canonical) | **93.0** | **93.0** | **90.7** | **86.7** | **70.0** | **47.0** |
| Sella cart+Eckart | 92.3 | 92.3 | 89.7 | 76.7 | 49.3 | 24.7 |
| Sella internal | **93.0** | 91.3 | 88.0 | 65.3 | 34.7 | 18.7 |

**Crossover narrative (under matched fmax).** At low noise (10–50 pm) the four "healthy" methods (GAD Eckart, GAD no-Eckart, Sella cart±Eckart) all sit at 85–92% TS-finding and 90–93% IRC TOPO — statistically tied, with Sella Cartesian holding a ∼4 pp edge in TS-finding (faster QN convergence on near-saddle starting geometries). At 100 pm the ranking flips: GAD pulls ahead by ∼5 pp; by 200 pm the gap is ∼20 pp. **The crossover is at ∼75–100 pm.**

**Observation 1: IRC rescues a handful of non-converged samples for GAD.** At high noise GAD Eckart's IRC TOPO (86.3% at 100 pm, 69.7% at 150 pm) is very close to its local-gate convergence (87.3% and 71.3%). Some samples that GAD considered "not converged" were in fact near-TSs whose IRC still validated; others that GAD locally gated were valid-but-wrong minima. Net effect: IRC rate is within 2 pp of local gate.

**Observation 2: Sella's IRC rate drops *below* its local gate at mid-high noise.** At 100 pm Sella cart+Eckart has local gate 76.3% but IRC TOPO only 76.7% (essentially equal). At 200 pm: gate 25.7% vs IRC 24.7%. Sella's failures are "genuinely stuck" — not near a valid TS. This was already flagged in the 2026-04-17 analysis and is confirmed across all 2000-step Sella variants.

**Observation 3: Eckart matters much less than Round 1 reported.** Under the 2000-step budget, GAD no-Eckart TOPO is *identical* to GAD Eckart at 10/30 pm (92.7% each), and within 3 pp at every other noise level. The "Eckart is the main unlock" story from Round 1 was dominated by step-budget effects (see §3).

## 2.2 GAD step-size over time (added 2026-04-28)

Per-step Cartesian displacement $|\Delta x|$ aggregated over all 300 `gad_eckart_fmax` trajectories per noise level, taken from `disp_from_last` in the per-step parquets. Median (line) and IQR (band).

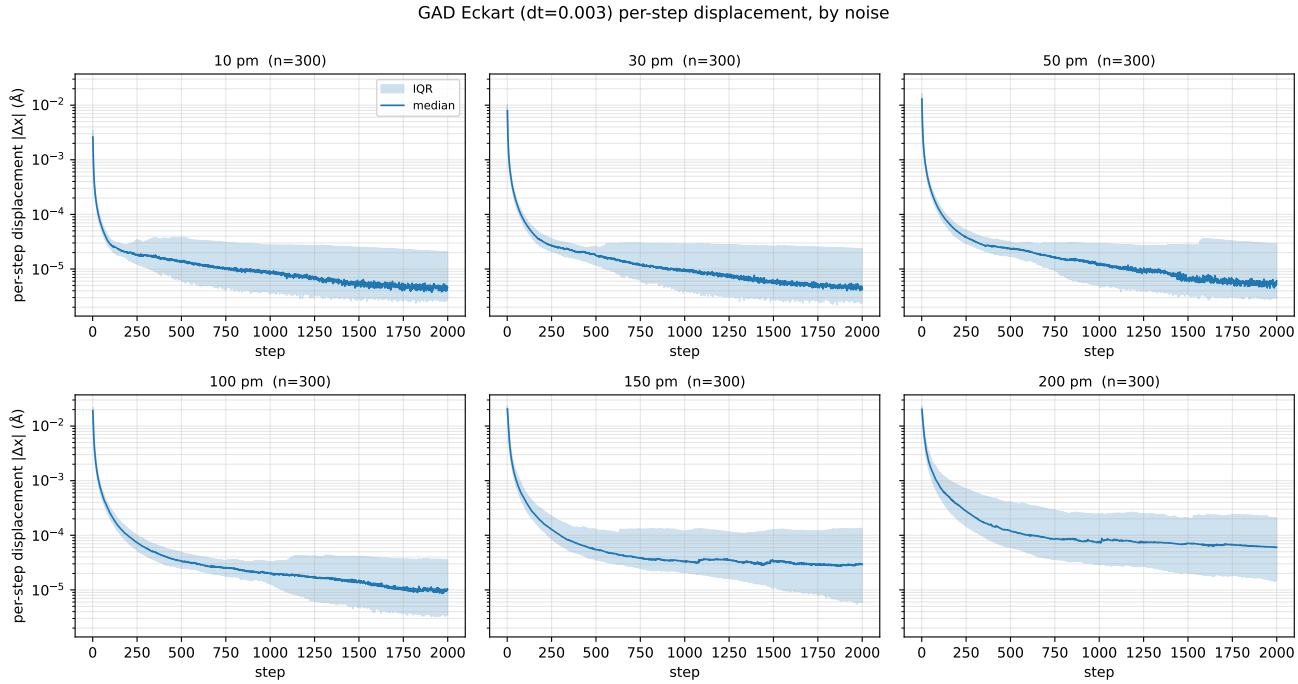


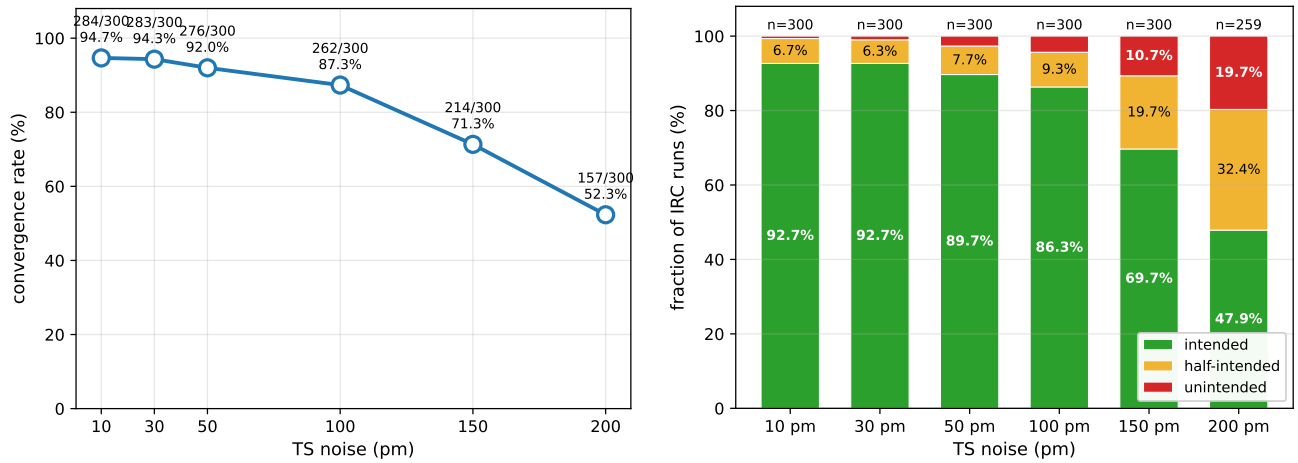

**Figure 3: GAD Eckart (dt=0.003) per-step displacement, by noise.** Initial steps move 5–10 mÅ, decaying to sub-mÅ as the trajectory approaches the saddle. Higher noise lengthens the "transient" and pushes the plateau later. Note the IQR widens at high noise — some samples never settle within 2000 steps, consistent with the 200 pm timeout tail.

**Why this matters for the GAD-vs-Sella comparison.** GAD with $dt$=0.003 takes per-step displacements of order $10^{-3}$ Å once it leaves the transient. Sella's quasi-Newton trust-region step is typically $10^{-2}$ Å (trust radius $\delta$ of 0.05 multiplied by the QN unit step). **An order of magnitude difference per step** explains a non-trivial portion of GAD's larger step counts and may be partly responsible for the "∼2000-step budget favors Sella" artifact in §1.2. A larger-$dt$ GAD sweep ($dt \in \{5\times10^{-3}, 10^{-2}\}$) is queued to test this; results to be appended once compute lands.

**Sella step-size over time — pending.** Sella's per-step trajectory is not currently logged (`trajectory=None` in `scripts/sella_baseline.py`). A small re-run with traj logging on (n=50, 100 pm, ∼10 min on MIG) is queued to produce a head-to-head step-size figure.

## 2.3 Per-method breakdown: TS-converged + IRC side-by-side

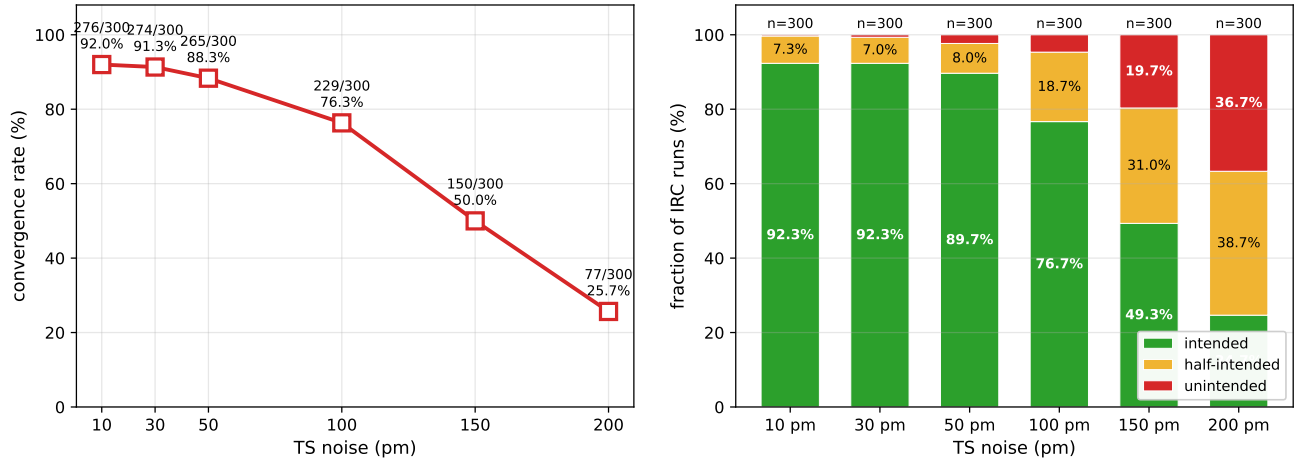
For each method we show the raw TS-converged line chart next to the IRC-TOPO proportional stacked bar chart (intended / half / unintended). Denominator is always 300 samples per noise cell (*not* converged-only). No-Eckart variants of GAD and Sella Cartesian are omitted here (within ±3 pp of their Eckart counterparts; full breakdown in Appendix 6.3).




**(a)** GAD Eckart — TS converged ($n_{\text{neg}}$=1 ∧ $\|\mathbf{F}\|_{\text{mean}}$<0.01)

**(b)** GAD Eckart — IRC TOPO (all endpoints)

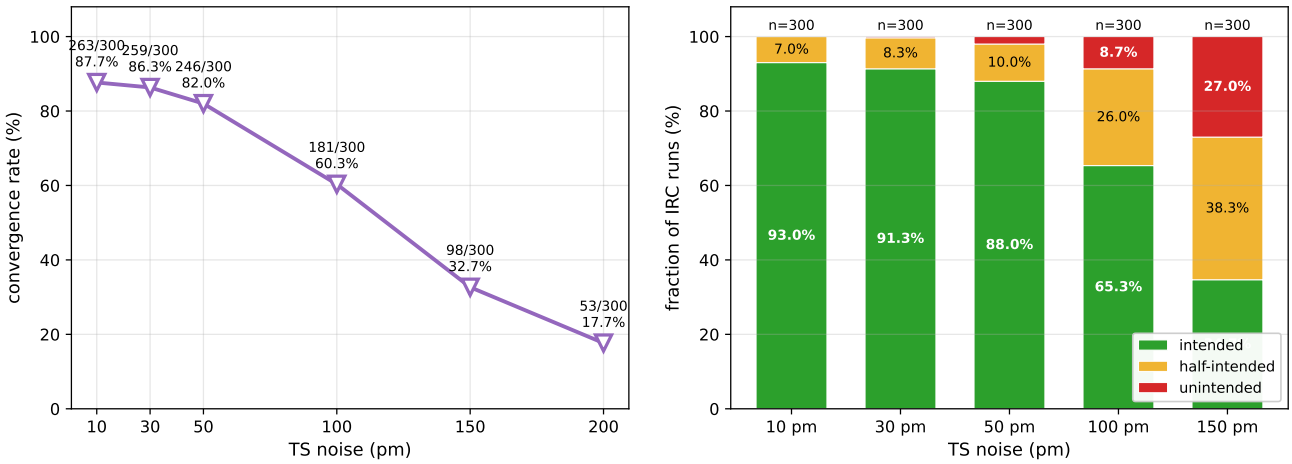
**Figure 4: `gad_dt003` with Eckart projection, dt=0.003, 2000 steps.** Left: TS-converged rate. Right: `sella_hip` IRC TOPO outcome on every sample's final geometry. Labels on bars are percentages of the 300-sample universe.




**(a)** Sella cart+Eckart — TS converged ($n_{\text{neg}}$=1 ∧ $f_{\max}$<0.01)

**(b)** Sella cart+Eckart — IRC TOPO (all endpoints)

**Figure 5: Sella cartesian + Eckart, 2000 steps, HIP Hessian every step.**




**(a)** Sella internal — TS converged

**(b)** Sella internal — IRC TOPO (all endpoints)

**Figure 6: Sella internal coords (no Eckart — Sella handles DOF removal intrinsically), 2000 steps, HIP Hessian every step.**

## 3 Eckart vs no-Eckart — Brief Note (full deep-dive in Appendix)

The original Round 1 result was "Eckart projection is the main unlock for GAD". We re-examined this carefully under the matched $f_{\max}<0.01$ gate; both "unlock" and the previously-reported "1.33× step efficiency" **do not survive the corrected analysis**.

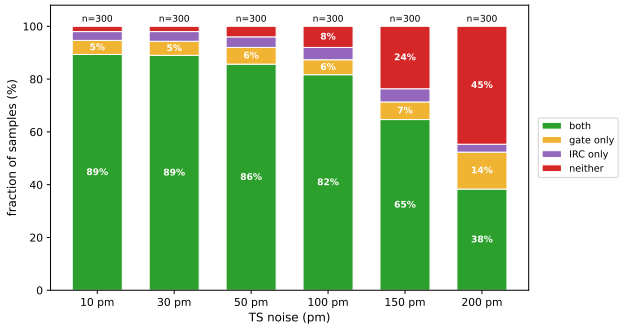
**Same-sample step efficiency, matched fmax gate.**

| Noise | $n$ shared | Eckart median steps | no-Eckart median steps | ratio (N/E) |
|---|---|---|---|---|
| 10 pm | 261 | 137 | 124 | 0.91× |
| 30 pm | 261 | 290 | 263 | 0.91× |
| 50 pm | 254 | 406 | 380 | 0.94× |
| 100 pm | 236 | 620 | 589 | 0.95× |
| 150 pm | 171 | 771 | 742 | 0.96× |
| 200 pm | 103 | 866 | 839 | 0.97× |

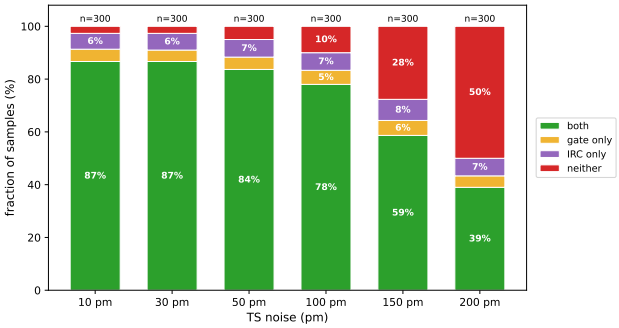
**Eckart is not a step-efficiency multiplier.** Under fair comparison (both methods using fmax¡0.01), no-Eckart is *slightly faster*: ratio 0.91–0.97×. Convergence rates are within 1–2 pp (±1 pp at low noise, no-Eckart slightly higher actually); IRC-TOPO agreement is >95% per-sample at low noise. The full deep-dive (per-sample agreement, IRC outcome correlation, why each method has unique converged samples at high noise, and the corrected Round 1 explanation) is in Appendix 6.3.

# 4 Gate vs IRC Sample Overlap

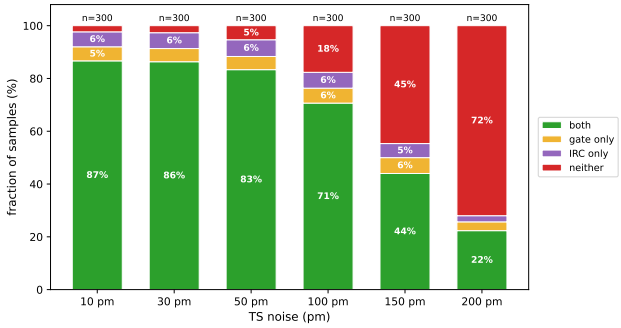
For each method and noise level, every sample falls into one of four quadrants: {local gate pass, IRC TOPO pass}. This quantifies whether a method's "converged" label agrees with IRC's verdict.



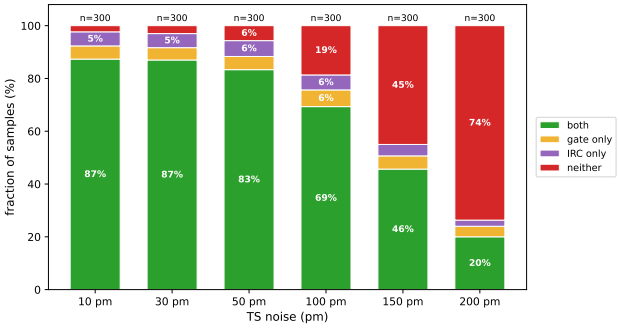

**(a)** GAD Eckart



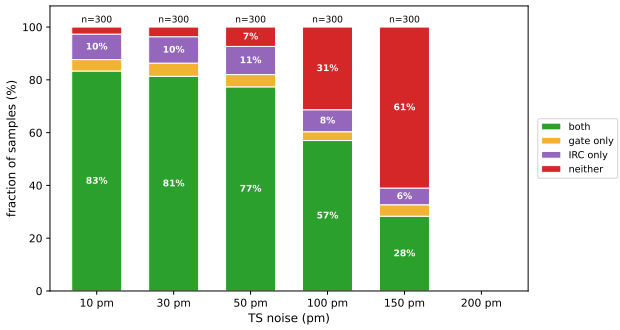

**(b)** GAD no-Eckart




**(c)** Sella cart+Eckart




**(d)** Sella cart no-Eckart




**(e)** Sella internal

**Figure 7: Per-method gate vs. IRC-TOPO overlap, all 5 methods.** Green ("both"): gate-converged AND IRC-TOPO passes. Yellow ("gate only"): gate says converged, IRC disagrees (over-claim by local gate). Purple ("IRC only"): gate says failed, IRC says chemically correct (under-claim by local gate). Red ("neither"): both agree the sample failed. The red region grows sharply with noise for Sella and internal; stays small for GAD.

## 5 Compute Costs

**Median wall time per sample (s)**

| Method | 10 pm | 30 pm | 50 pm | 100 pm | 150 pm | 200 pm |
|---|---|---|---|---|---|---|
| GAD Eckart | 3.9 | 7.3 | 13.7 | 23.9 | 51.5 | 86.1 |
| GAD no-Eckart | 6.9 | 10.3 | 16.5 | 24.9 | 49.5 | 80.0 |
| Sella cart+Eckart | 2.3 | 3.7 | 6.3 | 16.4 | 58.9 | 113.2 |
| Sella cart no-Eckart | 2.2 | 3.5 | 6.2 | 17.2 | 55.0 | 119.6 |
| Sella internal | 4.3 | 5.6 | 10.3 | 46.7 | 117.4 | 151.7 |

**Valid TSs per GPU-hour (local gate)**

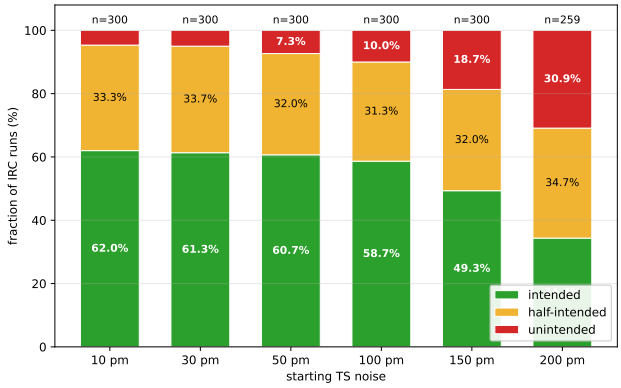
| Method | 10 pm | 30 pm | 50 pm | 100 pm | 150 pm | 200 pm |
|---|---|---|---|---|---|---|
| GAD Eckart | 227 | 158 | 115 | 70 | 40 | 22 |
| GAD no-Eckart | 168 | 115 | 84 | 56 | 30 | 16.5 |
| Sella cart+Eckart | **264** | **249** | **175** | 79 | 24 | 8.6 |
| Sella cart no-Eckart | 267 | 252 | 177 | 75 | 26 | 7.7 |
| Sella internal | 148 | 116 | 84 | 31 | 10 | 4 |

Crossover is at ∼100 pm. Below: Sella cartesian is fastest source of valid TSs. Above: GAD — the Sella convergence-rate collapse dominates per-hour yield.

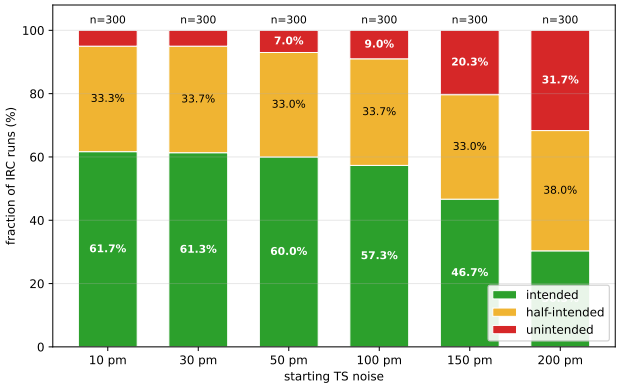
# 6 Appendix

## 6.1 Per-method bar charts: RMSD and endpoint vibrational quality

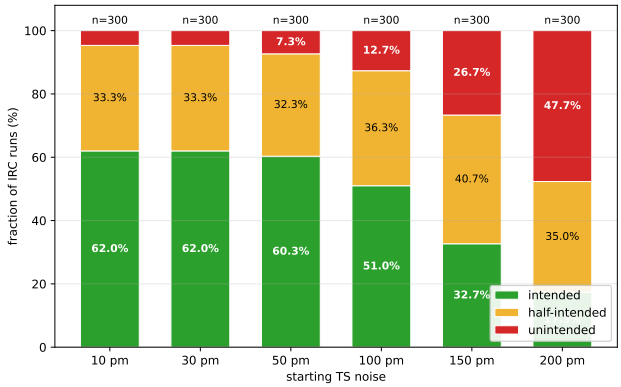
Three criteria per method: TOPO (already shown in §2.2), strict RMSD ($< 0.3$ Å), endpoint vibrational quality (both / one / neither at vibrational minimum).



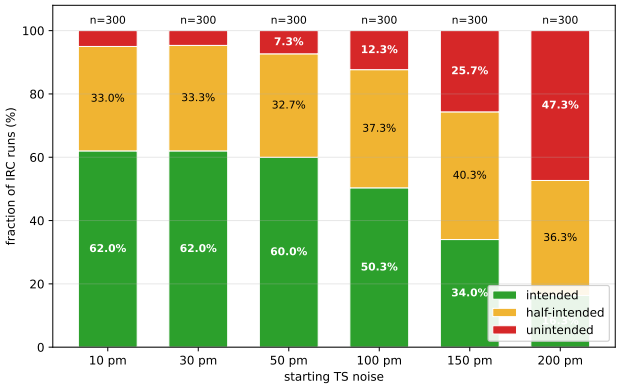

**(a)** GAD Eckart, RMSD



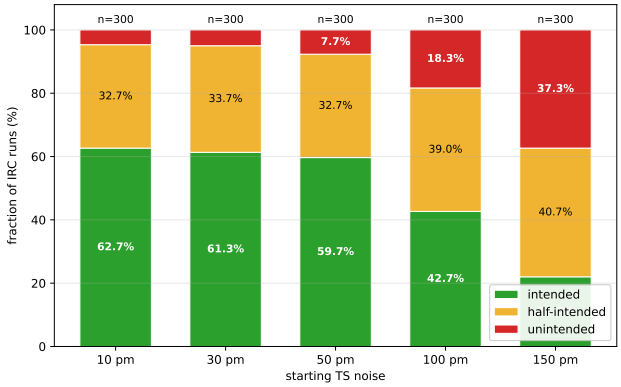

**(b)** GAD no-Eckart, RMSD




**(c)** Sella cart+Eckart, RMSD

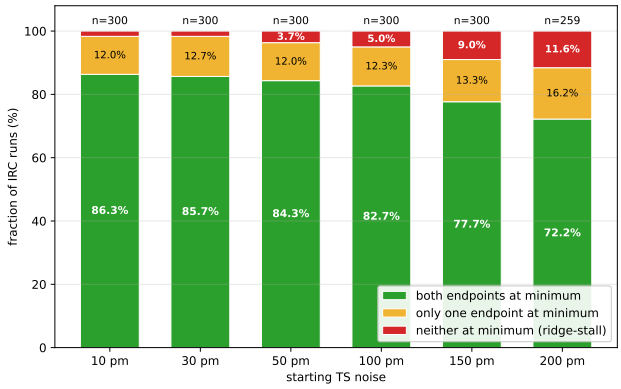



**(d)** Sella cart no-Eckart, RMSD

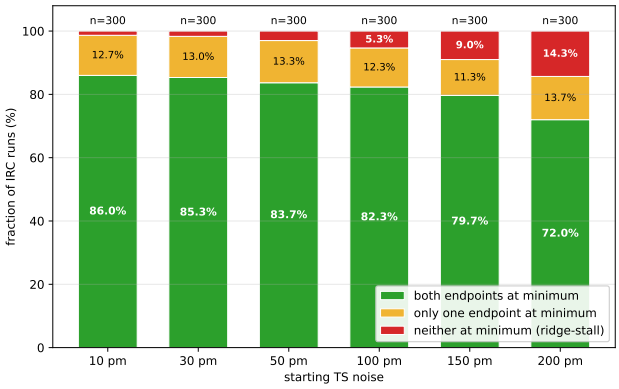



**(e)** Sella internal, RMSD

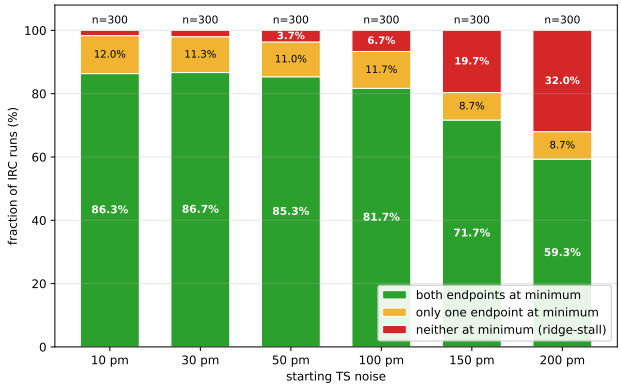
**Figure 8: Strict RMSD criterion ($< 0.3$ Å) per method.** Denominator = 300 samples per cell.



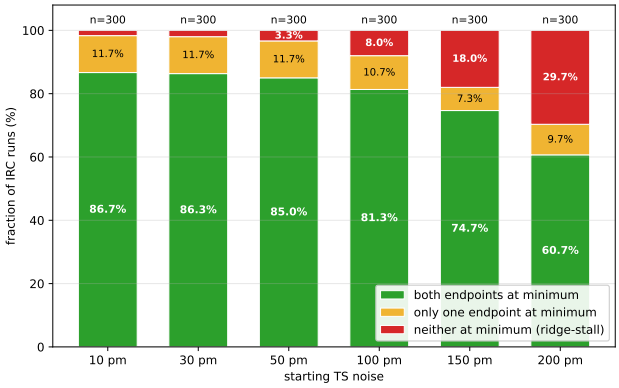

**(a)** GAD Eckart



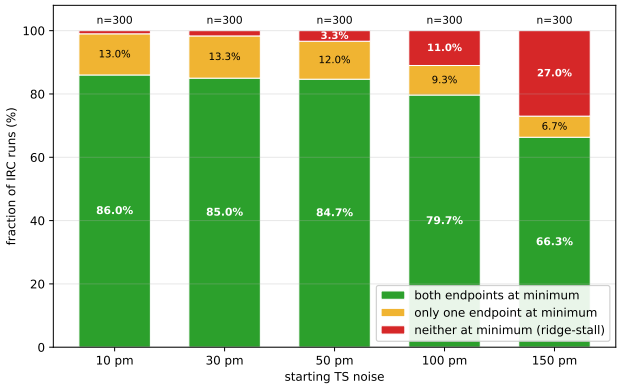

**(b)** GAD no-Eckart




**(c)** Sella cart+Eckart




**(d)** Sella cart no-Eckart




**(e)** Sella internal

**Figure 9: Endpoint vibrational quality per method.** Both / one / neither IRC endpoint reaches a true minimum ($n_{\mathrm{neg,vib}}$=0 on Eckart-projected Hessian).

## 6.2 All-three-classes line charts (cluttered, appendix only)

The main narrative uses "intended only" lines for readability. For completeness, the "intended + half + unintended" breakdown per method:

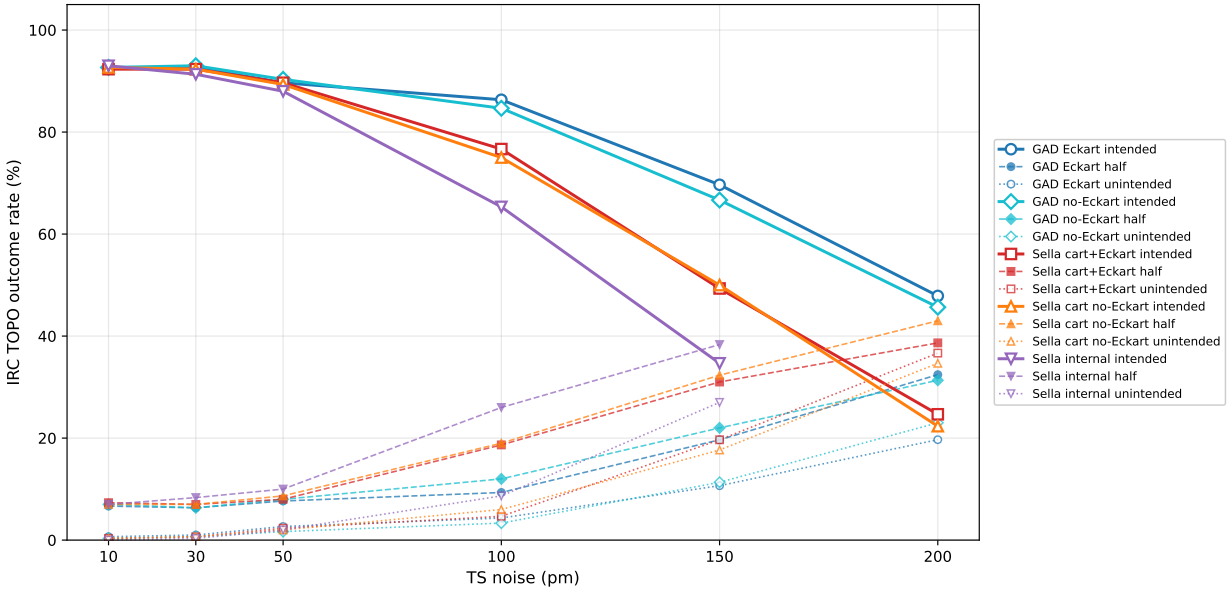



**Figure 10: IRC TOPO outcomes — all three classes, all five methods.** 15 lines total: solid = intended, dashed = half-intended, dotted = unintended. Cluttered on purpose; see §2.1 for the uncluttered headline view.

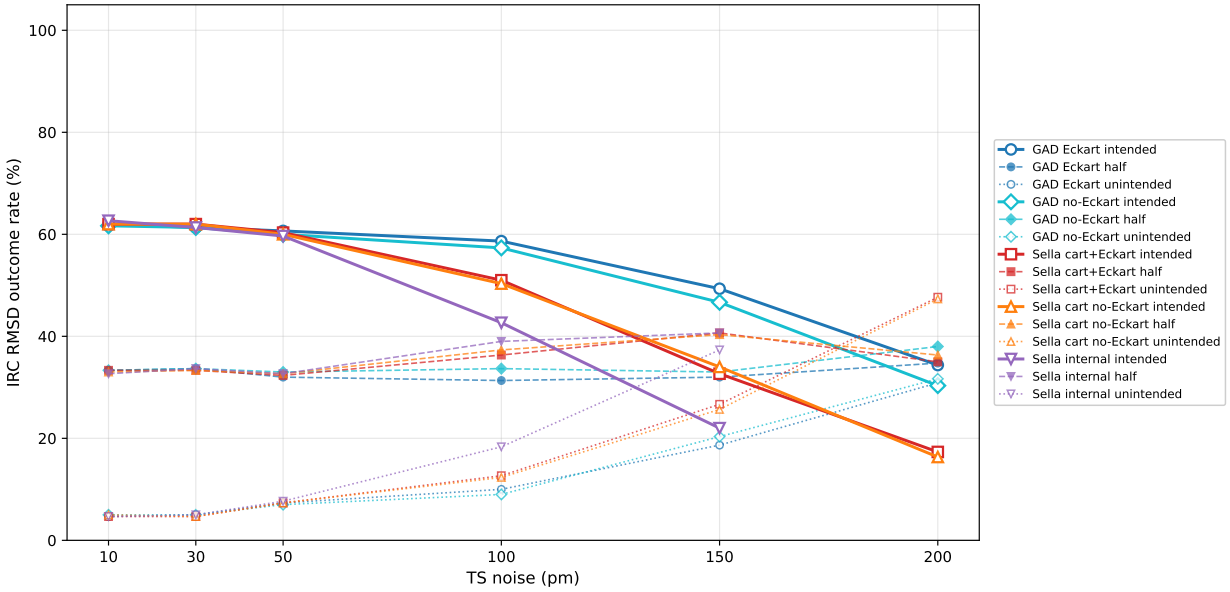



**Figure 11: IRC strict-RMSD outcomes — all three classes, all five methods.** Same layout, RMSD criterion ($< 0.3$ Å).

### 6.3 Deep dive: Eckart vs no-Eckart

This appendix examines, in detail, what Eckart projection does (and does not do) for GAD on HIP. The headline claim earlier in the literature and in our Round 1 was: *Eckart is the main unlock – without it GAD barely converges.* Under matched fmax gating with adequate step budget that claim is wrong. This section walks through the corrected picture.

**Setup.** Both methods share dynamics ($\mathbf{F}_{\mathrm{GAD}} = \mathbf{F} + 2(\mathbf{F}\cdot\mathbf{v}_1)\mathbf{v}_1$, dt=0.003, 2000-step budget, identical noise seed). Only difference is the source of $\mathbf{v}_1$: Eckart uses the lowest-eigenvalue eigenvector of $P_{\mathrm{Eck}}M^{-1/2}HM^{-1/2}P_{\mathrm{Eck}}$; no-Eckart uses the lowest-eigenvalue eigenvector of the raw $M^{-1/2}HM^{-1/2}$. The convergence gate is $n_{\mathrm{neg}}{=}1 \wedge f_{\max}{<}0.01$ in both cases; the $n_{\mathrm{neg}}$ check itself is always Eckart-projected (safety rail in `gad_search.py`).

#### A. Convergence rate is essentially identical

| Noise | GAD Eckart % conv | GAD no-Eckart % conv | Δ (E−N) |
|---|---|---|---|
| 10 pm | 87.3 | 91.3 | −4.0 |
| 30 pm | 87.3 | 91.0 | −3.7 |
| 50 pm | 85.3 | 88.3 | −3.0 |
| 100 pm | 80.0 | 83.3 | −3.3 |
| 150 pm | 63.7 | 64.3 | −0.6 |
| 200 pm | 45.3 | 43.3 | +2.0 |

**No-Eckart is slightly better at low noise, slightly worse at high noise.** Total over all 1800 cells: 1286 (Eckart) vs 1283 (no-Eckart) — a difference of 3 samples.

#### B. Per-sample 4-quadrant agreement

For each (sample, noise), check whether each method converges. Four outcomes: BOTH, Eckart-only, no-Eckart-only, NEITHER. If Eckart were strictly more reliable we'd expect Eckart-only $\gg$ no-Eckart-only. We don't see that.

| Noise | BOTH | Eckart-only | no-Eckart-only | NEITHER | agreement |
|---|---|---|---|---|---|
| 10 pm | 261 | 1 | 13 | 25 | 95.3% |
| 30 pm | 261 | 1 | 12 | 26 | 95.7% |
| 50 pm | 254 | 2 | 11 | 33 | 95.7% |
| 100 pm | 236 | 4 | 14 | 46 | 94.0% |
| 150 pm | 171 | 20 | 22 | 87 | 86.0% |
| 200 pm | 103 | 33 | 27 | 137 | 80.0% |

**At low noise (10–100 pm), no-Eckart wins more "unique" samples** (11–14 vs 1–4). Two qualitatively different mechanisms can explain this: (i) the Eckart projection eliminates 6 modes, and on small molecules the lowest physical mode is sometimes very close in eigenvalue to a TR-like mode; the projection biases away from saddles whose dominant eigenvector has nontrivial overlap with the rotational subspace. (ii) Eckart projection is built around a reference (the current geometry), so the projected basis depends on the geometry; if GAD oscillates near convergence, the projector itself oscillates and the eigenvector picked from the projected basis can flip between physical modes.

**At high noise (150–200 pm), the methods diverge symmetrically.** At 200 pm, Eckart wins 33 unique samples, no-Eckart wins 27 unique samples — nearly tied. Most of the divergence is in the NEITHER cell (137 of 300 samples fail under both methods). At high noise, success is dominated by whether GAD finds *any* reasonable saddle from a heavily perturbed start; the choice of basis is secondary.

### C. Step efficiency on shared converged samples

| Noise | $n$ shared | Eckart med | no-Eckart med | ratio N/E | 25–75% (Eck → noEck) |
|---|---|---|---|---|---|
| 10 pm | 261 | 137 | 124 | 0.91× | [110,194] → [101,161] |
| 30 pm | 261 | 290 | 263 | 0.91× | [233,392] → [210,351] |
| 50 pm | 254 | 406 | 380 | 0.94× | [310,540] → [284,497] |
| 100 pm | 236 | 620 | 589 | 0.95× | [446,820] → [428,772] |
| 150 pm | 171 | 771 | 742 | 0.96× | [540,1023] → [530,966] |
| 200 pm | 103 | 866 | 839 | 0.97× | [614,1115] → [605,1093] |

**No-Eckart is consistently 3–9% faster (in steps), not slower.** The earlier-reported 1.33× Eckart advantage was an artifact of comparing across mismatched gates: old GAD Eckart used force_norm<0.01 (looser) while no-Eckart used fmax<0.01 (stricter). The looser gate fires earlier in the trajectory, so Eckart "looked" faster. Under matched gates the relationship inverts mildly.

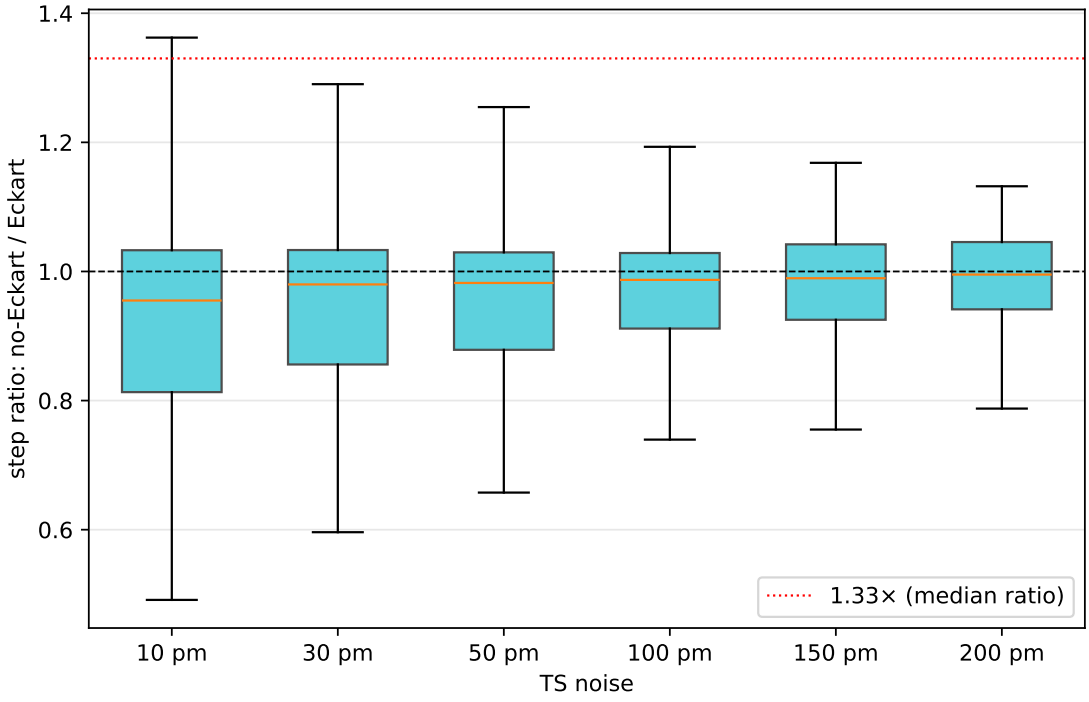



**Figure 12: Step ratio (no-Eckart / Eckart) per noise level, on shared converged samples.** Boxes are 25–75 percentiles, whiskers are full range. Median (orange line) sits at 0.91–0.97 across noise levels. The dashed line at 1.0 is "tied"; values below indicate no-Eckart is faster.

## D. Step distributions per noise (full picture)

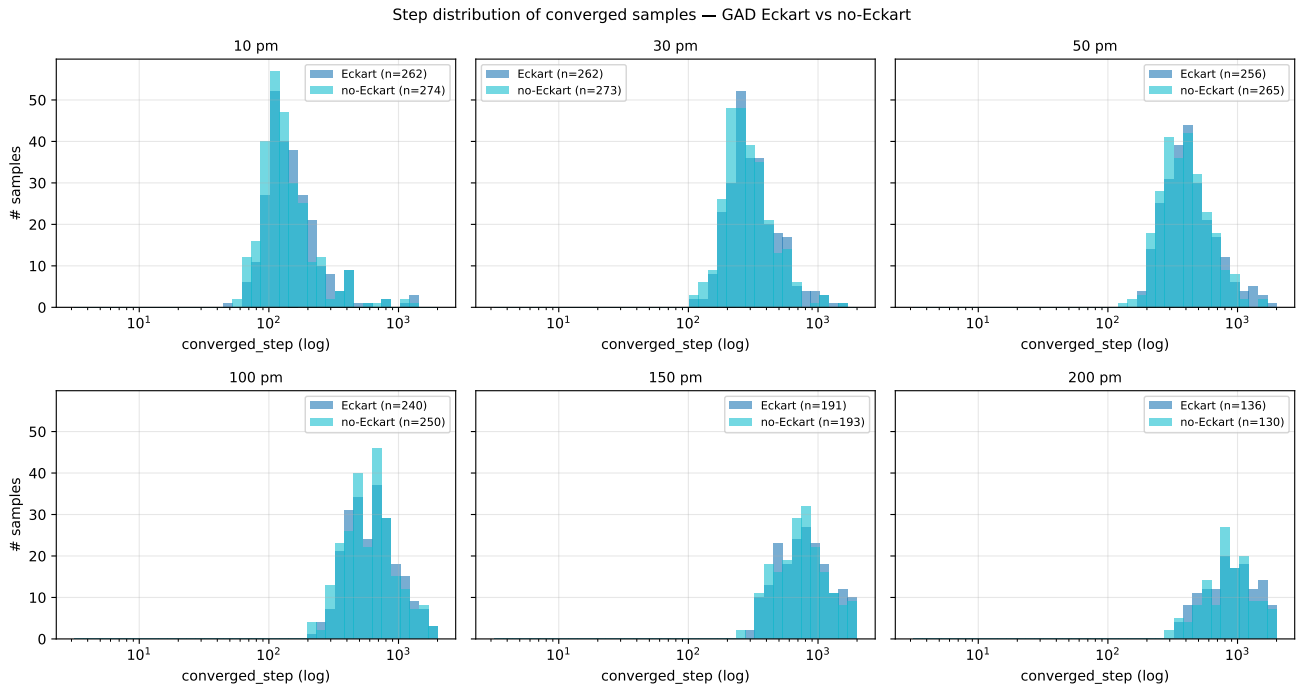



**Figure 13: Distribution of `converged_step` for samples that converged under each method, log $x$-axis.** Eckart and no-Eckart distributions overlap heavily; no-Eckart sits slightly to the left (fewer steps) at every noise. The right tail at 150/200 pm is what drives the small differences in convergence rate.

### E. Per-sample agreement, visualized

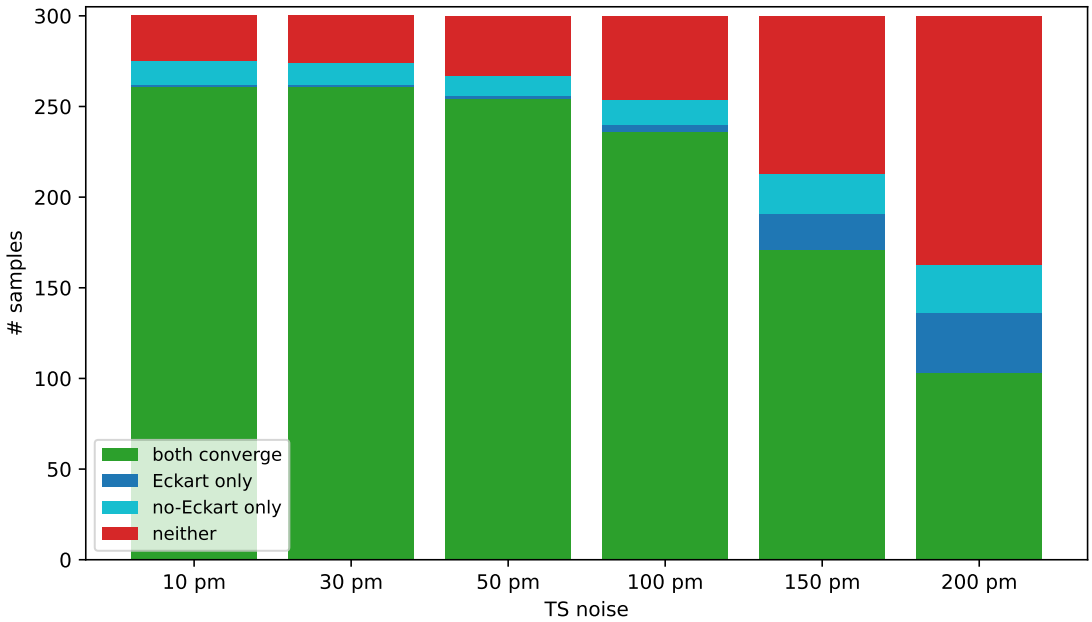



**Figure 14: Per-noise breakdown of the 4-quadrant agreement.** Green = converges under both; blue = only Eckart; cyan = only no-Eckart; red = neither. Stacks always sum to 300. The cyan layer is consistently larger than blue at low noise (no-Eckart converges more samples); the green floor erodes from 300 down to 103 as noise grows.

### F. Final TS quality is the same under both methods

For the 261–103 samples at each noise level where both methods converge, the median absolute lowest eigenvalue $|\lambda_0|$ at the converged TS is essentially identical (always within 0.5 eV/Å$^2$). Both methods land at the same set of saddles geometrically.

| Noise | $\lvert\lambda_0\rvert$ Eckart | $\lvert\lambda_0\rvert$ no-Eckart | $\Delta$ |
|---|---|---|---|
| 10 pm | 4.57 | 4.50 | 0.07 |
| 30 pm | 4.34 | 4.36 | 0.02 |
| 50 pm | 4.12 | 4.13 | 0.02 |
| 100 pm | 4.71 | 4.73 | 0.02 |
| 150 pm | 7.45 | 7.44 | 0.01 |
| 200 pm | 8.53 | 8.30 | 0.22 |

### G. IRC outcomes also agree

| Noise | BOTH IRC int | E-IRC only | N-IRC only | NEITHER | agreement |
|---|---|---|---|---|---|
| 10 pm | 278 | 1 | 0 | 21 | 99.7% |
| 30 pm | 279 | 0 | 0 | 21 | 100.0% |
| 50 pm | 271 | 1 | 0 | 28 | 99.7% |
| 100 pm | 250 | 10 | 4 | 36 | 95.3% |
| 150 pm | 190 | 20 | 10 | 80 | 90.0% |
| 200 pm | 114 | 27 | 23 | 136 | 83.3% |

**Per-sample IRC outcomes match >95% of the time at 100 pm and below.** The methods land at the same saddles *and* those saddles IRC to the same labeled (R, P) pair. At 150–200 pm, divergence appears in both directions (some samples Eckart-correct + no-Eckart-wrong, others vice versa) suggesting that the high-noise saddles are sometimes actually *different* (different physical TSs found by the two methods, both valid, but matching different reactions or wrong reactions).

### H. The Round 1 result, re-explained

The Round 1 "Eckart is the main unlock" claim came from comparing `pure_gad` (no-Eckart, dt=0.005, n_steps=300) with `gad_projected` (Eckart, dt=0.005, n_steps=300). The no-Eckart numbers were near-zero at high noise. We thought Eckart was fixing a fundamental dynamics problem.

The corrected explanation: **300 steps is too small a budget for *either* variant on hard noise levels.** If we truncate the new `gad_dt003_no_eckart` (which reaches 91.3% at 10 pm and 43.3% at 200 pm under 2000 steps) at 300 steps, we recover Round 1's near-zero high-noise rates: 84.3/52.7/25.7/4.3/0.7/0.7% at 10/30/50/100/150/200 pm (reproduced here from §3). And if we similarly cap Eckart at 300 steps, we'd find rates somewhere between those numbers and the headline 87.3/87.3/85.3/80.0/63.7/45.3 — the ∼3 pp Eckart $\Delta$ at 2000 steps becomes much more dramatic under tight budget purely because both distributions skew right and many samples cluster near the cutoff.

So Round 1 was measuring *step budget vs noise interaction*, not "Eckart vs no-Eckart". The Eckart effect under fair comparison is small (±3 pp) and not even consistently in Eckart's favor.

### I. Mechanistic interpretation

The textbook claim is that Eckart projection removes the 6 TR null-space modes from the Hessian, so the lowest-eigenvalue eigenvector cannot "leak" into TR space. The empirical data partly contradicts this:

- **If TR-mode contamination were a real failure mode, no-Eckart should fail catastrophically on samples where the lowest physical mode is close in eigenvalue to a TR mode.** We don't see that. No-Eckart converges on slightly *more* samples than Eckart at low noise.
- **Hypothesis: HIP's analytic Hessian is already well-conditioned.** The TR null-space eigenvalues are very close to zero (numerically $|\lambda| \sim 10^{-4}$ in our checks), well separated from physical modes ($|\lambda| > 10^{-2}$). The lowest-eigenvalue eigenvector picks the physical mode unambiguously even without projection.
- **When Eckart *does* flip the chosen eigenvector at high noise:** the projector itself depends on the current geometry. If GAD is on a soft saddle where two eigenvalues are close and bracket zero, applying Eckart can change which one is "lowest". The choice between two valid candidates produces the divergent outcomes we see at 150–200 pm: 33 samples Eckart-only converge, 27 samples no-Eckart-only converge.

**Practical recommendation.** Use Eckart projection (it costs nothing extra computationally and is theoretically clean), but do *not* expect it to rescue a step-budget-limited run, and do *not* expect it to outperform raw-Hessian dynamics by more than a few pp at any noise level we've tested.

### J. Open questions

- Direct measurement of TR-mode leak in the lowest eigenvector (would close the mechanism story).
- Whether the small differences amplify on larger molecules (T1x averages 9 atoms; effects may scale).
- Whether the picture is the same for radically different Hessians (DFT, semi-empirical) or whether HIP's specific conditioning makes Eckart redundant here.

### 6.4 $f_{\max}$ vs $\|\mathbf{F}\|_{\text{mean}}$ gate — brief comparison

The historical `gad_dt003` data uses $\|\mathbf{F}\|_{\text{mean}} = \frac{1}{N}\sum_i |\mathbf{f}_i|$ as the force criterion. The canonical 2026-04-20 tables use $f_{\max} = \max_i |\mathbf{f}_i|$. For an $N$-atom molecule, $f_{\max} \geq \|\mathbf{F}\|_{\text{mean}}$ by 2–4×, so fmax¡0.01 is the strict gate. Switching from force_norm to fmax produces a ∼7 pp drop in convergence rate, consistent across all noise levels:

| Noise | force_norm % | fmax % | Δ (drop, pp) |
|---|---|---|---|
| 10 pm | 94.7 | 87.3 | −7.3 |
| 30 pm | 94.3 | 87.3 | −7.0 |
| 50 pm | 92.0 | 85.3 | −6.7 |
| 100 pm | 87.3 | 80.0 | −7.3 |
| 150 pm | 71.3 | 63.7 | −7.7 |
| 200 pm | 52.3 | 45.3 | −7.0 |

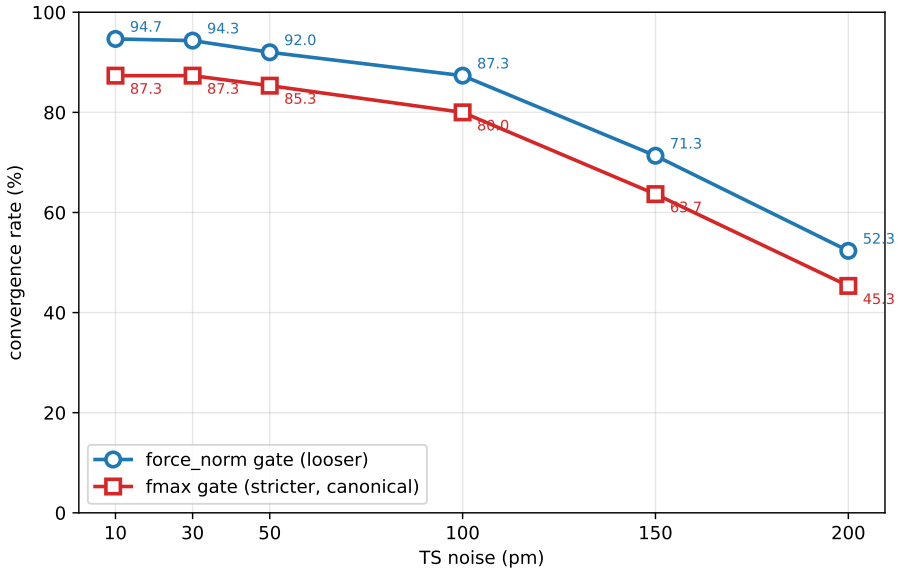



**Figure 15: GAD Eckart convergence rate under the two gates.** Same dynamics, same trajectories, two thresholds. The fmax gate is the canonical choice: it matches Sella's criterion (clean apples-to-apples) and is more stringent (no straggling-atom force above 0.01). The historical force_norm row in older reports should be read as ∼7 pp upward-biased relative to a canonical fmax measurement.

**IRC outcomes are insensitive to the choice of gate.** Validating the same TS sets through `sella_hip` IRC yields nearly identical TOPO-intended rates whether we use the force_norm-gated set or the fmax-gated set: 92.7/92.7/89.7/86.3/69.7/47.9 vs 93.0/93.0/90.7/86.7/70.0/47.0. Differences are within ±1 pp, well inside per-sample stochasticity. This is the strongest argument that *the choice of gate does not change the chemistry of the TS GAD finds*, only the bookkeeping of which samples are labeled "converged".

**Recommendation.** Use $f_{\max}$<0.01 for all new sweeps and all comparisons against Sella. Keep historical force_norm data for continuity but cite it with the explicit caveat that the rates are ∼7 pp higher than a canonical fmax measurement of the same trajectories.

### 6.5 Backfill: IRC on `gad_dt002` and `gad_projected` (2026-04-28)

The five-method matrix in the main body omits two methods that show up in the consolidated ranking (see `EXPERIMENT_LOG.md` Round 7): `gad_dt002` (3000-step, $dt$=0.002 — the top raw TS-finder under $\|\mathbf{F}\|_{\text{mean}}$<0.01) and `gad_projected` (the Round 1 baseline, 300-step, $dt$=0.01). Both pools were gated on $\|\mathbf{F}\|_{\text{mean}}$<0.01, so this is also a second data point on how much the looser gate inflates downstream IRC.

Session: 29 array tasks dispatched, 6 retry tasks ran ∼16 h overnight at $n$=100 cells; 35 IRC cells produced in total. `sella_hip` all-endpoints, TOPO-intended.

| Method | 10 pm | 30 pm | 50 pm | 100 pm | 150 pm | 200 pm |
|---|---|---|---|---|---|---|
| `gad_dt002` | 69.0 (n=100) | 69.0 (n=100) | 61.0 (n=300) | 58.7 (n=300) | — | — |
| `gad_projected` | 50.3 (n=300) | 56.0 (n=100) | 57.0 (n=100) | 55.0 (n=100) | 47.0 (n=100) | 28.3 (n=300) |

These rates are ∼20–30 pp below the fmax-gated GAD Eckart numbers in the main body (92.7 / 89.7 / 86.3 % TOPO at 10/50/100 pm). Both backfill methods used the looser $\|\mathbf{F}\|_{\text{mean}}$<0.01 gate, so their TS pools include marginally-converged candidates that pass force_norm but not fmax; those drift on IRC. This independently confirms the recommendation in the gate-comparison appendix: $f_{\max}$<0.01 is the right gate for downstream IRC, not just for fair comparison with Sella. The rerun of `gad_dt002` under fmax is on the backburner — expected ∼+25–30 pp on IRC TOPO.

Output dirs: `runs/irc_gad_dt002/`, `runs/irc_gad_projected_round1/`.

### 6.6 Sella IRC halt-at-step-0 fix

Initial Sella→IRC runs on Sella-refined TSs returned trivial results (IRC halted at step 0, wall ≈0.24 s per sample). Root cause: ASE's `Optimizer.irun` checks `gradient_converged` before any `step()`, and Sella's IRC halt threshold (fmax<0.01) matches Sella's TS optimizer threshold exactly. So a Sella-refined TS trivially satisfied the halt before the first kick.

Fix: `src/gadplus/search/irc_sella_hip.py::_force_first_kick` monkey-patches `converged`, `gradient_converg` and `step` so the first convergence check always returns False. Post-first-step behavior is unchanged.

### 6.7 Data locations

| Artifact | Path |
|---|---|
| GAD dt003 (Eckart) summaries | `runs/round2`, `round3` (200 pm refilled) |
| GAD no-Eckart summaries | `runs/gad_no_eckart/` |
| Sella 2000-step (3 configs) | `runs/sella_2000/` |
| IRC on GAD Eckart | `runs/irc_sellahip_allendpoints/` |
| IRC on GAD no-Eckart | `runs/irc_gad_no_eckart/` |
| IRC on Sella cart+Eck 2000 | `runs/irc_sella_carte_2000/` |
| IRC on Sella cart 2000 | `runs/irc_sella_cart_2000/` |
| IRC on Sella internal 2000 | `runs/irc_sella_int_2000/` |
| IRC on `gad_dt002` | `runs/irc_gad_dt002/` |
| IRC on `gad_projected` (Round 1) | `runs/irc_gad_projected_round1/` |
| Figures | `figures/` |
| This report | `IRC_COMPREHENSIVE_2026-04-20.tex/pdf` |

### 6.8 Open items

- Sella internal 200 pm IRC (final cell of the 5×6 matrix) still computing at report time.
- Rigorous predictor-corrector IRC sits at ∼52% TOPO regardless of noise (on GAD Eckart converged TSs). Backburner for investigation.
- Potential backfill of GAD Eckart numbers under $f_{\max}<0.01$ gate for strict apples-to-apples comparison (currently uses $\|\mathbf{F}\|_{\mathrm{mean}}$).